

Erdős Problem #1013

EP-1013

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

Let $h_3(k)$ be the minimal n such that there exists a triangle-free graph on n vertices with chromatic number k . Find an asymptotic for $h_3(k)$, and also prove $\lim_{k \rightarrow \infty} \frac{h_3(k+1)}{h_3(k)} = 1$.

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>